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EXP:06 DELUGE SCRIPTS USING ZOHO CREATOR

(INTERNSHIP MANAGEMENT SYSTEM – 10 PROGRAMS WITH OUTPUT)

1. Aim

To implement and document Deluge scripts in Zoho Creator for automating common operations in the Internship Management System, including application ID generation, approval status updates, internship duration calculation, email notification, and attendance working-hour calculation.

2. Procedure

1. Open the Internship Management System application in Zoho Creator.
2. Navigate to Workflow and choose the relevant form and record event.
3. Open the Deluge script editor and enter the required script.
4. Save and enable the workflow or action.
5. Submit or edit a test record with suitable values.
6. Verify the generated ID, calculated values, updated status, or email output.
7. Capture the script and result screenshots for documentation.

3. Deluge Programs with Actual Screenshots and Output

Program 1: Automatic Internship Application ID

Aim: To automatically generate a unique Application ID when a new internship application is submitted.

Actual Deluge Script Screenshot:



Figure 3.1: Program 1: Automatic Internship Application ID – screenshot from Zoho Creator

Expected / Observed Output: Example output: APP-20261018184411 (the timestamp will vary for each submission).

Note: Actual screenshot supplied by the user from the Zoho Creator Deluge editor.

Program 2: Application Approval Workflow

Aim: To update the application approval status and application status when the Application Approved field is set to true.

Actual Deluge Script Screenshot:



Figure 3.2: Program 2: Application Approval Workflow – screenshot from Zoho Creator

Expected / Observed Output: When Application Approved = true, Approval Status becomes Approved and Application Status becomes Selected.

Note: Actual screenshot supplied by the user from the Zoho Creator Deluge editor.

Program 3: Calculate Internship Duration

Aim: To calculate internship duration in days using the start date and end date.

Actual Deluge Script Screenshot:



Figure 3.3: Program 3: Calculate Internship Duration – screenshot from Zoho Creator

Expected / Observed Output: For Start Date = 01-Nov-2026 and End Date = 01-Dec-2026, the calculated duration is 30 days.

Note: Actual screenshot supplied by the user from the Zoho Creator Deluge editor.

Program 4: Student Approval Email Notification

Aim: To send an email notification to the student when the application is approved.

Actual Deluge Script Screenshot:

```
sendmail
[
    from :"nethrachandragandhi.t.2024.cse@rajalakshmi.edu.in"
    to :"nethracg07@gmail.com"
    subject :"Internship Application Approved"
    message :"Dear Student, your internship application has been approved. Please log in to the Internship Management System to view your application details."
]
```

Figure 3.4: Program 4: Student Approval Email Notification – screenshot from Zoho Creator

Expected / Observed Output: A message with the subject “Internship Application Approved” is sent to the configured recipient. The uploaded script shows a fixed test recipient.

Note: Actual screenshot supplied by the user from the Zoho Creator Deluge editor. The recipient shown is a fixed test address; replace it with a student's email field if the notification must go to each applicant dynamically.

Program 5: Calculate Attendance Working Hours

Aim: To calculate attendance working hours from check-in and check-out times.

Actual Deluge Script Screenshot:

```
if(input.Check_In_Time != null && input.Check_Out_Time != null)
{
    checkin_seconds = input.Check_In_Time.getHour() * 3600 + input.Check_In_Time.getMinutes() * 60 + input.Check_In_Time.getSeconds();
    checkout_seconds = input.Check_Out_Time.getHour() * 3600 + input.Check_Out_Time.getMinutes() * 60 + input.Check_Out_Time.getSeconds();
    if(checkout_seconds >= checkin_seconds)
    {
        input.Working_Hours = ((checkout_seconds - checkin_seconds) / 3600.0).round(2);
    }
    else
    {
        input.Working_Hours = 0;
    }
}
```

Figure 3.5: Program 5: Calculate Attendance Working Hours – screenshot from Zoho Creator

Expected / Observed Output: For Check-In = 09:00 and Check-Out = 15:00, Working Hours = 6.00. If check-out is earlier than check-in, the script sets working hours to 0.

Note: Actual screenshot supplied by the user from the Zoho Creator Deluge editor.

4. Additional Deluge Examples (Programs 6–10)

The following five examples are included to meet the ten-program requirement. They are additional examples, not represented as screenshots or as already deployed workflows. Configure and test each one in a compatible form event before using it in the application.

6. Check Application Selection

Program Code:

```
if(input.Application_Status == "Selected")
{
    info "Application selected.";
}
else
{
    info "Application is not selected yet.";
}
```

Sample Output: For Application Status = Selected, the execution log displays “Application selected.”

7. Validate Internship Date Range

Program Code:

```
if(input.Start_Date != null && input.End_Date != null)
{
    if(input.End_Date < input.Start_Date)
    {
        alert "End Date must be on or after Start Date.";
        cancel submit;
    }
}
```

Sample Output: If End Date is earlier than Start Date, submission is cancelled with a validation message.

8. Check Available Positions

Program Code:

```
if(input.Available_Positions != null)
{
    if(input.Available_Positions > 0)
    {
        info "Internship positions are available.";
    }
    else
    {
        info "No internship positions are currently available.";
    }
}
```

Sample Output: For Available Positions = 3, the execution log displays “Internship positions are available.”

9. Student Registration Confirmation

Program Code:

```
info "Student registration submitted successfully.";
```

Sample Output: The execution log displays “Student registration submitted successfully.”

10. Evaluate Overall Rating

Program Code:

```
if(input.Overall_Rating >= 4)
{
    input.Performance_Result = "Excellent";
}
else
{
    input.Performance_Result = "Needs Improvement";
}
```

Sample Output: For Overall Rating = 4.6, Performance Result becomes “Excellent.” This example requires compatible field types and a suitable workflow event.

5. Result

Five actual screenshots from the Zoho Creator Deluge editor have been inserted for the scripts supplied by the user. Five additional Deluge examples with sample outputs are included to complete the ten-program experiment. The application-related scripts demonstrate ID generation, status updates, duration calculation, email notification, and attendance-hour calculation.

6. Conclusion

The experiment demonstrates how Deluge scripting supports workflow automation in Zoho Creator. Actual script screenshots provide evidence of the configured project features, while the additional examples illustrate further conditional logic and validation possibilities.